F005 v0.5.0-p2

Raw Acquisition Provenance & Initial-State Qualification

22 October 2026 | Evidence-first qualification protocol

Classification: DEVELOPMENT / E0_REVIEW_NOT_CERTIFIED. This patch is a read-only source and initial-state audit; it does not create a new physical SST/Euler result.

1. Problem and input boundary

The prior p1 audit checked 3,400 development TRAIN trajectories. Its two relative-polarity hypotheses gave smaller TRAIN-only RMSE for +1 at horizons 0.1, 0.3 and 0.5 s. That is a data-dependent relative sign, *not* a physical verification of the two raw coordinate frames. The reported first-sample magnitude was small but cannot certify stationary internal fluid flow.

The thesis supplies geometry $L = 0.080$ m, $\ell = 0.031$ m, $h = 0.043$ m and nominal model density $\rho = 998$ kg m^{-3}. It documents subtracting a pre-motion optical baseline based on 1,000 recorded samples. The exported NPY data do not in themselves provide the original pre-motion frames.

2. What p2 does and does not access

Reads	Boundaries
Locked v0.4.3 split CSV	SHA-256 locked, 3,400 TRAIN and 850 VALIDATION IDs; zero overlaps in the registered proxy-family graph. No measured heights read.
Existing geometry JSON	Compared to documented thesis values; never overwritten.
Historical p1 JSON	Quoted as TRAIN-only descriptive evidence; never refits or reselects sign.
Acquisition source text	Only from explicit evidence folder, static search and SHA-256; no external code executed.
Small Zenodo archive	Only the short `load_dataset.py` member, if available; loader is not raw acquisition proof.
Raw pre-motion CSVs	Only files deliberately mapped to locked TRAIN IDs; no TEST/OOD, DEVELOPMENT VALIDATION, or large `DATASET.zip`.

3. Central identifiability limit

For a linear response operator $\mathcal{L}$,

$$\eta_{\mathrm{obs}}(t) = s_\eta \, \mathcal{L}[s_a a_{\mathrm{raw}}](t), \qquad s_a, s_\eta \in \{-1, +1\}.$$

Only $s_a s_\eta$ is identified by the measured scalar input/output pair, not the two signs separately. The original code and coordinate-frame chain are required for independent axis verification.

4. Initial-state evidence and falsifiable checks

For each genuinely available original pre-motion record, the audit checks its SHA-256, its registered development TRAIN ID, time monotonicity, the declared actuation start, and the count N_{pre}. It records the scalar baseline scatter

$$\sigma_{\eta,\mathrm{pre}} = \sqrt{\frac{1}{N_{\mathrm{pre}}} \sum_{j=1}^{N_{\mathrm{pre}}} (\eta_j - \overline{\eta})^2}$$

and the least-squares trend of $\eta(t)$ before actuation. $N_{\mathrm{pre}} \geq 1000$ implements the documented optical baseline method as a *screen*; it is not a guarantee of zero vorticity, zero velocity, or equilibrated modal motion.

$$\eta(0) \approx 0, \qquad \dot{\eta}(0) \approx 0 \quad \not\Rightarrow \quad \mathbf{v}(\mathbf{x}, 0) = \mathbf{0}.$$

The audit never sets `initial_equilibrium_confirmed=true` automatically.

5. Gate table

Gate	Requirements and status logic
P2_LOCKED_SPLIT	Exact source hash, valid proxy group and size counts. Fails closed if different.
P2_GEOMETRY	Source-exact dimensions, model constants and sensor $x = L/2$ verified.
P2_INPUT_AXIS	Original exporter and robot axis chain, line-level citation, reviewer semantics.
P2_HEIGHT_AXIS	Camera rotation, image coordinate origin, sign transformations and wall mapping traced.
P2_PREMOTION	Original timestamped pre-motion traces and start-event provenance, per trajectory.
P2_HYDRO_REST	Independent physical evidence/protocol for quiescent fluid; not deducible from scalar signal.
E0_MAIN_EULER	**BLOCKED** pending explicit evidence review. The audit is unable to self-certify.
E5 / E7 / E8	Euler experiment NOT RUN; PKLSA NOT RUN; SST claims UNRESOLVED.

6. Reproduction

Unzip into the unchanged F005 root. Run `run_all_v050_p2.cmd TEST` then `run_all_v050_p2.cmd`. Review output JSON and markdown summary. Put any original exporter/recording scripts and raw pre-motion traces under `sources/acquisition_evidence/`; use the independent `P2_EVIDENCE_MANIFEST.json` file to cite their exact line ranges. Preserved historical outputs are not rewritten.

7. Explicit limitations

The robot dataset may store commanded rather than directly measured acceleration. The source transformation from command frames to the NPY array and the camera optical sign convention must be traced rather than inferred from a low RMSE. A corrected sign alone is insufficient if the dynamic start state is unverified. No nonlinear Euler, vorticity reconstruction or SST ontology is tested.

References

References

[1] D. Camilotto (2026), *A Comprehensive Data-Driven Framework for Robotic Liquid Sloshing Control: From Extensive Experimental Model Validation and Enhancement to Trajectory Optimization*, Master's thesis, Politecnico di Milano. `https://hdl.handle.net/10589/253216`.

[2] N. Giulietti et al. (2026), *Robot-executed longitudinal sloshing dataset for rectangular tank liquid handling*, Zenodo. `https://doi.org/10.5281/zenodo.20039757`.

[3] O. M. Faltinsen and A. N. Timokha (2009), *Sloshing*, Cambridge University Press, ISBN 9780521881111.